



Associative learning and representativeness

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Why look beyond rational expectations?



- ▶ Every model of expectations is a cognitive model
- ▶ The cognitive critique: *Standard economic models are inconsistent with any possible model of cognition.*
- ▶ Financial economist answer: it's sufficient to have some very smart wealthy people with powerful computers
 - ▶ Maybe not. At least for the hardest problems.
- ▶ Average excess return of U.S. stocks over bonds, last 100 years: 5–7 percentage points a year
- ▶ If beliefs satisfied rational expectations, it is highly unlikely that we would have experienced these returns

How might a cognitive psychologist think about this?



- ▶ They would ask for evidence that investors
 - ▶ knew that there was an equity premium
 - ▶ believed that disasters were possible
 - ▶ viewed the equity premium as compensation for disasters
- ▶ Evidence points to the opposite
- ▶ To know there was a premium, an investor would have had to guess the equilibrium

Without statistical analysis, investors must rely on their memory.



1. Non-standard utility — technically still rational expectations
2. Ad hoc forecast rules: adaptive, extrapolative, constant-gain — no account of why that functional form
3. Beliefs tied to cognitive foundations: **Bayesian with a flawed database**

The program is class 3.

There is already a theory



- ▶ Retrieved-context theory (Howard & Kahana 2002; CMR and its successors)
- ▶ Built to fit recall and recognition data from the laboratory
- ▶ A century of constraint, accumulated outside economics, on parameters economics would otherwise leave free

The experimental paradigms



Two phases: study, then test. The stimuli are words.

Test	The task	Isolates
free recall	recall the list, any order	retrieval dynamics: contiguity, primacy, recency
cued recall	study $A-B$; see A , recall B	single associations
serial recall	recall in order	order memory
item recognition	studied, or new?	memory match
associative recognition	$A-B$ yes; $A-D$ no	associations without recall

A century of measurement on the same tasks — a common framework to make progress.



We don't have perfect memory. Why? Three possibilities:

1. We don't "see" it — lossy input function
2. We forget it
3. **It's still there — we just can't find it**
 - ▶ evidence: sometimes we find it, even when we don't want to

How do you find something that's lost? Need a path to get there.

- ▶ Similarity
- ▶ Recency
- ▶ Temporal contiguity

But also: competition.



1. Most actions are choices we do not notice: an associative engine supplies the next action
2. Some involve deliberation — imagining the future
3. Deliberation involves free recall: a laboratory paradigm and a real-world activity
4. Free recall is measurably context-dependent; the choices are too
5. Recall creates a new copy, counted for later recall: beliefs can differ from the physical process in a first-order way



- ▶ Recall reads out frequency: the more often seen, the more probable (Estes 1976; Wachter & Kahana 2024)
- ▶ Representativeness is a **reversal**: the more frequent answer loses. A frequency read-out cannot deliver one.
- ▶ Associative recognition asks “do these go together?” It matches the contexts the cue and the target retrieve. The target arrives with its whole associative history.

Claim: the Kahneman–Tversky judgments are acts of associative recognition. Representativeness then follows as a theorem.



Kahneman & Tversky (1973). Three groups, one judgment each about nine graduate fields: base rate, similarity of the description to the field, likelihood that Tom is in the field.

Tom W. is of high intelligence, although lacking in true creativity. He has a need for order and clarity, and for neat and tidy systems in which every detail finds its appropriate place. His writing is rather dull and mechanical, occasionally enlivened by somewhat corny puns and by flashes of imagination of the sci-fi type. . . . He seems to have little feel and little sympathy for other people and does not enjoy interacting with others.

The sketch matches the stereotype of a computer scientist, one of the rarer fields.

Tom W.: likelihood tracks similarity, runs against base rates



Field	Base rate (%)	Similarity (rank)	Likelihood (rank)
Computer science	7	2.1	2.5
Engineering	9	2.9	2.6
Humanities	20	7.2	7.6
Social science	17	8.2	8.0

Rankings: 1 = most similar, most likely. KT's categories:
humanities & education; social science & social work.

- ▶ Likelihood rankings: near-perfect correlation with similarity, negative correlation with base rates.
- ▶ Bayes must weight by the base rate:
$$P(\text{field} \mid \text{Tom}) \propto P(\text{Tom} \mid \text{field}) P(\text{field}).$$
- ▶ Representativeness: judge $P(\text{field} \mid \text{Tom})$ by resemblance to the stereotype.

Example 2: a controlled version (BCGSS 2021)

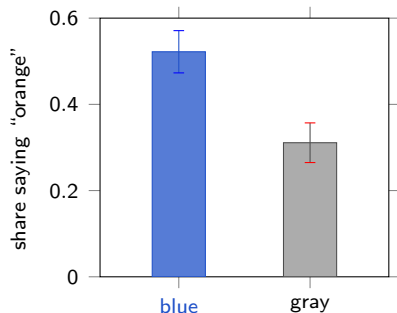


Bordalo, Coffman, Gennaioli, Schwerter, & Shleifer, *Psych. Review*. Subjects watch a stream of images; the treatments differ only in 25 of them:

- ▶ Blue: 10 orange numbers, 15 blue numbers, 25 blue words
- ▶ Gray: 10 orange numbers, 15 blue numbers, 25 gray shapes

What is the likely color of a randomly drawn NUMBER?

- ▶ More say orange in the blue treatment than in gray.
- ▶ The orange share falls as blue words are replaced by orange words.



Study 1: 52.2% vs. 31.1%, $N=790$, 95% CI.

The two experiments, and the reversal



	Notation	Tom W.	BCGSS
Cue	d	Tom W.	Number
Target 1	h	Computer science	Orange
Target 2	$-h$	Humanities	Blue
Decoy	$-d$	Other people	Words

d = data (the cue); h = hypothesis (a target).

- ▶ Among numbers, blue outnumbers orange, 15 to 10. Subjects say orange.
- ▶ The judgment reverses the true ordering. An account must explain a flip, not degraded accuracy.
- ▶ Bayes ranks targets by $P(h \mid d) \propto P(d \mid h) P(h)$.

Definition (Representativeness heuristic)

Cued with d , the agent judges h more likely than $-h$, even though $-h$ is in fact more likely.



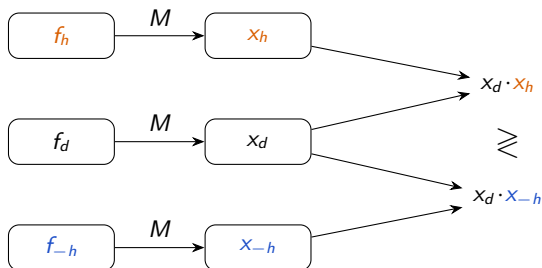
- ▶ Features f are aspects of the environment (basis vectors). A cue retrieves a unit-length *input* context bound to its features: $x^{\text{in}} \propto Mf$, $\|x^{\text{in}}\| = 1$.
- ▶ Studying drifts context and accumulates associations:

$$x_t = (1 - \zeta) x_{t-1} + \zeta x_t^{\text{in}}, \quad M_t = M_{t-1} + x_t f_t^\top.$$

- ▶ Retrieved context is a weighted tally of where the cue's features co-occurred:

$$x^{\text{in}} \propto M_0 f_d + \sum_s x_s (f_s^\top f_d).$$

associative recognition: CMR-IA, Jin, Halpern, Wachter & Kahana



- ▶ Cue and each candidate target retrieve a context through M . Items studied together share a context, so their retrieved contexts align.
- ▶ Judge h more likely than $-h \iff x_d \cdot x_h > x_d \cdot x_{-h}$.
- ▶ Contexts are unit length: the comparison carries a normalization, a consequence of the model rather than an assumption.



A worked example: the study list

Each item is a bundle of features; each lays down a context x :

NUMBER		WORD		BLUE		ORANGE
$\updownarrow$		$\updownarrow$		$\updownarrow$		$\updownarrow$
$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$		$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$		$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$		$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$
Blue '5'	//	Orange '3'	//	Blue 'CAT'		
$\underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}}$		$\underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}}$		$\underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}}$		
x_1		x_2		x_3		

A worked example: matrix and comparison



Study list: Blue '5' // Orange '3' // Blue 'CAT'

$$M = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \quad \text{cols: (NUM, WORD, BLUE, ORANGE)}$$

Retrieved contexts $x_j \propto Mf_j$:

$$x_{\text{NUMBER}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad x_{\text{BLUE}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad x_{\text{ORANGE}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Recognition match = cosine of retrieved contexts:

$$x_{\text{NUMBER}} \cdot x_{\text{ORANGE}} = \frac{1}{\sqrt{2}} \approx 0.71 > x_{\text{NUMBER}} \cdot x_{\text{BLUE}} = \frac{1}{2} = 0.50.$$

Orange wins: the blue word pulls x_{BLUE} off the number context.



Orthonormal contexts and items in population proportions give the match a closed form:

$$x_h \cdot x_d = \frac{P(h, d)}{\sqrt{P(h) P(d)}}.$$

Blue treatment:

$$x_{\text{ORANGE}} \cdot x_{\text{NUMBER}} = \frac{P(\text{ORANGE}, \text{NUMBER})}{\sqrt{P(\text{ORANGE}) P(\text{NUMBER})}} = \frac{10}{\sqrt{10} \sqrt{25}} = 0.63$$

$$x_{\text{BLUE}} \cdot x_{\text{NUMBER}} = \frac{P(\text{BLUE}, \text{NUMBER})}{\sqrt{P(\text{BLUE}) P(\text{NUMBER})}} = \frac{15}{\sqrt{40} \sqrt{25}} = 0.47$$

- ▶ **Blue** appears 40 times (15 numbers + 25 words): $\sqrt{P(\text{BLUE})}$ penalizes it. **Orange** appears only as numbers.
- ▶ The square root comes from the geometry of unit-length contexts rather than from a fitted parameter.

Recall reads out the truth; recognition flips it



Task	Ranks by	Verdict
Recall (read-out through M^T)	$P(h, d)$	blue 15 > orange 10 ✓
Recognition (context match)	$P(h, d) / \sqrt{P(h)}$	orange 0.63 > blue 0.47 ✗

- ▶ Competition (the 25 blue words) lowers recall of every number. It never makes a blue number less recallable than an orange one.
- ▶ Cue dependence and competition are in every memory model. They move levels. The reversal requires the target's other associations, which enter only through the match.
- ▶ Reduced-form accounts capture the reversal by fitting an interference term; here it is derived.

The conjunction fallacy



Bill is “intelligent, but unimaginative, compulsive, and generally lifeless ... strong in mathematics but weak in social studies.” Subjects rank: accountant (A) > accountant who plays jazz (A&J) > jazz player (J).

$$\underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{x}_1} \quad // \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad // \quad \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

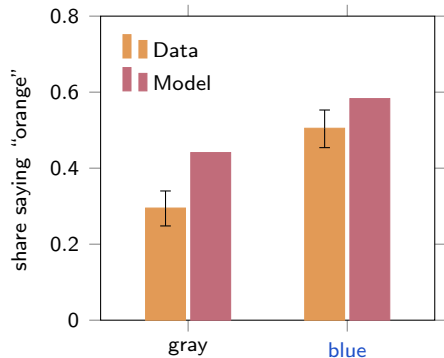
$\mathbf{x}_1 \qquad \qquad \mathbf{x}_2 \qquad \qquad \mathbf{x}_3$

Cue Bill; the conjunction is a composite target, $\mathbf{x}_{A \cap J} \propto M(f_A + f_J)$:

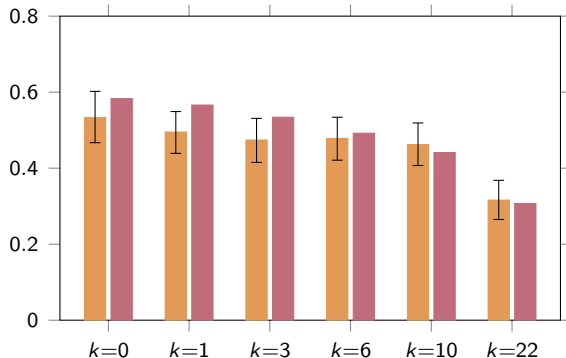
$$\mathbf{x}_{\text{BILL}} \cdot \mathbf{x}_A = \frac{1}{\sqrt{2}} > \mathbf{x}_{\text{BILL}} \cdot \mathbf{x}_{A \cap J} = \frac{1}{\sqrt{3}} > \mathbf{x}_{\text{BILL}} \cdot \mathbf{x}_J = 0.$$

A count of episodes preserves the subset order; a context match need not.

Model and data (one noise parameter)



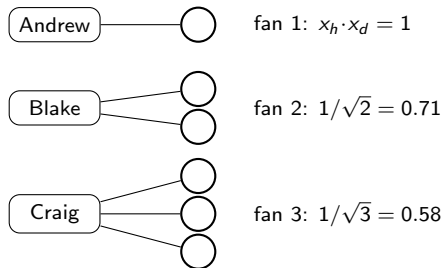
Study 1: blue vs. gray treatment



Study 2: k of the 25 blue words replaced by orange words

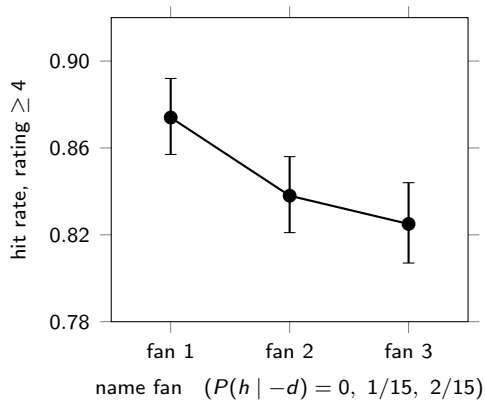


- ▶ Paradigm of Pantelis, van Vugt, Sekuler, Wilson & Kahana (2008): learn names for faces.
- ▶ 16 faces, 9 names. Every face has one name; a *name* goes with 1, 2, or 3 faces.
- ▶ $P(\text{name} \mid \text{face}) = 1$ in every condition. Only the name's other faces vary.
- ▶ Six study–test cycles; test shows a face with its name or another studied name; rate 1–6.
- ▶ 244 participants (Prolific).



Prediction: recognition of a studied pair falls as $1/\sqrt{f}$.

Frequency accounts: no target-fan effect, or a positive one through name frequency.



Participant-weighted means, 95% CI.

Studied pairs, regressed on model-implied similarity $1/\sqrt{f}$:

Outcome	β on $1/\sqrt{f}$	t
Hit, rating ≥ 4	0.114	5.2
Hit, rating = 6	0.242	8.0
Confidence (1–6)	0.640	7.1

Participant and cycle fixed effects; t clustered by participant.

- ▶ Fan 3 $\rightarrow$ fan 1: +4.8 points of recognition, +0.27 points of confidence.
- ▶ Discriminability (ROC area) falls with fan: 0.89, 0.86, 0.84.

The square root, and what the decoys rule out

NEW



Unrestricted fan dummies (fan 3 omitted). The model pins the ratio of the fan-1 to the fan-2 effect: $(1 - 1/\sqrt{3})/(1/\sqrt{2} - 1/\sqrt{3}) = 3.26$; linearity in $P(h \mid -d)$ pins it at 2.

Outcome	Fan 1	Fan 2	Ratio	p : ratio = 3.26	p : ratio = 2
Hit, rating ≥ 4	0.048	0.012	3.86	0.75	0.11
Hit, rating = 6	0.102	0.026	3.90	0.60	0.01
Confidence (1–6)	0.269	0.075	3.58	0.79	0.04

- ▶ The $1/\sqrt{f}$ spacing is never rejected; linearity in $P(h \mid -d)$ is rejected twice.
- ▶ Decoys (face with a wrong name): false alarms are *higher* for high-fan names (0.24 vs. 0.28), so the fan effect is not caution about names seen often.
- ▶ The model's literal decoy prediction, no fan effect on false alarms, holds at the strict criterion ($p = 0.19$) and fails at the lenient one ($p < 0.01$).

Where this shows up in markets



Role	BCGSS	Analyst expectations	Recession forecasts	Equity carve-outs
Cue d	Number	High past growth	Recession-like indicators	New-economy state
Decoy $-d$	Word	Low past growth	Normal indicators	Old-economy state
Target h	Orange	High future growth	Recession	Palm
Target $-h$	Blue	Low future growth	Normal times	3Com (owns Palm)

- ▶ Value premium: the many low-growth firms erode the match between high past growth and low future growth. High-past-growth firms are judged high-growth; prices are bid up; returns disappoint (La Porta 1996).
- ▶ Excess volatility: recession-like indicators evoke the recession context; a filter would not.
- ▶ Negative stubs (Lamont & Thaler 2003): 3Com pays off in both states, so its match with the new-economy context falls below Palm's, a claim it contains.



- ▶ Bordalo, Gennaioli, Shleifer and coauthors: the representativeness ratio $P(h \mid d)/P(h \mid -d)$ as the kernel of truth (BCGS 2016); exponentiated into diagnostic expectations (BGS 2018); a similarity function with a Luce rule fit to the experiments (BCGSS 2021); recall-based sampling with an interference term (BCGKS 2023).
- ▶ Bhatia (2017): similarity from word embeddings reproduces the judgments; a special case of a match in a vector space, with the space built by language rather than by the experiment.
- ▶ Busemeyer, Pothos, Franco & Trueblood (2011): the conjunction fallacy from sequential projection onto incompatible subspaces, which also predicts order effects. Here it comes from normalizing a composite target; the match is symmetric, so there is no order effect.
- ▶ Here: the functional form is derived (the square root), recognition rather than recall generates the conjunction fallacy, and the paired-associates laboratory disciplines the mechanism.

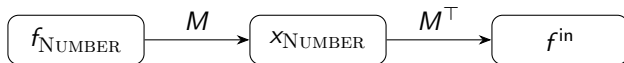


- ▶ Probabilistic judgment in the Kahneman–Tversky tasks is associative recognition: compare the contexts a cue and a target retrieve.
- ▶ Representativeness is then a theorem of retrieved-context theory:

$$x_h \cdot x_d = \frac{P(h, d)}{\sqrt{P(h) P(d)}}.$$

- ▶ The $\sqrt{P(h)}$ normalization delivers base-rate neglect and the orange/blue reversal; recall in the same model reads out the truth.
- ▶ A paradigm with every association built in the laboratory shows the target-fan effect with the predicted spacing.
- ▶ The mechanism speaks to the value premium, excess volatility, and negative stubs.

Backup: recall reads out the true frequency

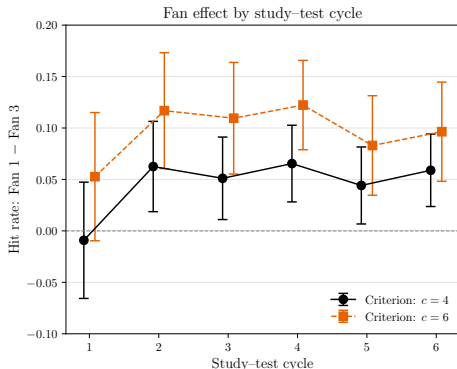


The cue retrieves the contexts of all numbers, then reads them back through M^T (contexts orthonormal):

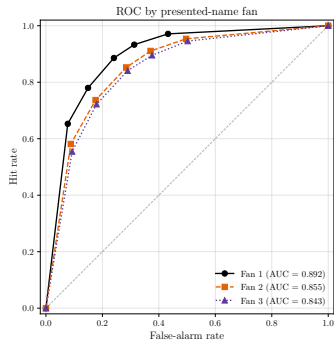
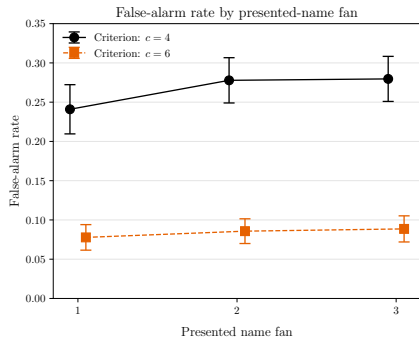
$$x_{\text{NUMBER}} \propto M f_{\text{NUMBER}} = \sum_{i \in \text{numbers}} x_i,$$
$$f^{\text{in}} = M^T x_{\text{NUMBER}} \propto \sum_{i \in \text{numbers}} f_i = 25 f_{\text{NUMBER}} + 15 f_{\text{BLUE}} + 10 f_{\text{ORANGE}}.$$

In coordinates (NUM, WORD, BLUE, ORANGE): $f^{\text{in}} \propto (25, 0, 15, 10) \Rightarrow \text{blue} > \text{orange}.$

Backup: fan effect by study–test cycle



- ▶ Fan-1 minus fan-3 hit rate, by cycle; participant fixed effects.
- ▶ Recognition improves with study: 73% in cycle 1 to 89% in cycle 6.
- ▶ The fan effect is absent in cycle 1 and stable from cycle 2 on (equality across cycles 2–6 not rejected, $p = 0.75$).



Left: false alarms by the fan of the wrong name shown. Right: ROC by presented-name fan; area 0.892, 0.855, 0.843.



- ▶ BCGSS (2021): a distorted probability measure,

$$\tilde{P}(h \mid d) \propto \frac{(c + P(h \mid d)P(d))^\alpha}{(c + P(h \mid -d)P(-d))^\gamma}.$$

- ▶ $c = 0$, $\alpha = 1 + \gamma$ gives diagnostic expectations:
 $\tilde{P}(h \mid d) \propto P(h \mid d) [P(h \mid d)/P(h \mid -d)]^\gamma.$

- ▶ Here:

$$\tilde{P}(h \mid d) \propto \frac{P(h \mid d)}{\sqrt{P(h \mid d) + P(h \mid -d)}},$$

with the exponent fixed by the geometry of context.